

MI&CT ESERVICE NOTE SN08110509b

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Subject: GOC6 Operator Console Service Introduction
For LightSpeed VCT (64 Slice Only) 7.1

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1. OBJECTIVE

This information has been provided to assist the GE Service Engineer and Service Operations groups with the market introduction of the LightSpeed VCT GOC6 Operator Console. The LightSpeed VCT (64) 7.1 GOC6 Operator Console replaces the EOL (end-of-life) LightSpeed VCT (64) 7.1 GOC5 Operator Console.

Note: VCT Select (32 slice system) is out of scope of this program



2. INTRODUCTION

The VCT GOC6 console is the equivalent replacement for the VCT GOC5 console. The VCT GOC6 console will be shipping in the fourth quarter of 2008 on all VCT 64 systems. The new console provides improved reliability, fault tolerance, addresses parts obsolescence and cost issues. The GOC6 console will support new (soon to be announced) capabilities and will soon be available for IB upgrades. VCT Select (Pro32) systems will not get VCT GOC6 consoles.

3. PRODUCT FEATURES

No new product features.

4. SITING REQUIREMENTS

For siting requirements, see Direction 5220253-1EN CT750 HD & VCT Pre-Installation Manual.

5. HARDWARE DIFFERENCES

- **DIG Computer:** combined DARC/Image Generator – diskless
 - **VeRB Computer:** Future Option: Volume Recon Box – diskless
 - **HSDA:** High Speed Disk Array – RAID 5 Controller
 - Standard broadband InSite connectivity
 - USB Service Key
- Note: With the introduction of the USB Service Key, it will not be able to be used for both MR and CT. This issue will be addressed in the future.**
- Ethernet Smart Switch – 16 ports 10/100/1000BaseT
 - **Operator Console Power Outlet Box - IEC Outlets:**
 - Circuit Breaker #1 – J12 thru J19
 - Circuit Breaker #2 – J2 thru J11 (J2 thru J5 external)
 - Circuit Breaker #3 - J21 and J22 (external)
 - **Gantry Reset Signal** no longer illuminates Gantry X-ray On indicator when Operator Console power is removed.

6. OPERATOR CONSOLE HARDWARE

6.1 HOST COMPUTER (xw8400)

CPU	Dual Intel Dual Core Xeon 2.00GHz 5130 Processors
	O/S 64-bit
Cache	4MB L2 cache
Memory	4GB DDR2-667MHz
RAM OS Disk	146GB, 15,000rpm 3.0Gb/s SAS HDD
Image Disk	146GB, 15,000rpm 3.0Gb/s SAS HDD
	250,000 uncompressed 512x512 images
Graphics Card	Nvidia Quadro FX1500 PCI Express 16x with 256MB Memory and Dual 400MHz RAMDA

6.2 DATA ACQUISITION AND IMAGE GENERATION COMPUTER (DIG)

CPU	Dual Intel Quad Core Xeon 2.33GHz 5410 Processor
	64-bit floating point data format
Cache	4MB L2 cache
RAM	16GB FBDIMM ECC PC5300
Graphics Card	nVidia® Tesla C870
SAS Host Adapter	LSI Corp LSISAS3801E
VDIP	GEHC PN 2382000

6.3 HIGH SPEED DISK ARRAY (RAID 5)

Scan Data Storage	8x 250GB SATA 7200rpm HDDs configured to 500 GB capacity
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6.4 VOLUME RECON BOX (VERB) (OPTIONAL)

CPU	Dual Intel Quad Core Xeon 2.33GHz 5410 Processor
	64-bit floating point data format
Cache	4MB L2 cache
RAM	16GB FBDIMM ECC PC5300
Graphics Card	nVidia® Tesla C870

6.5 SCIM AND INTERCOM

SCIM	Start Scan, Pause Scan, Stop Scan, Move to Scan, Stop Move, Prescribed Tilt and Emergency Stop. Intercom speaker, microphone and volume controls
ICOM	Auto Voice, Gantry Reset, X-ray Abort.

6.6 PERIPHERALS

- **Color LCD monitors** (2 standard):
 - 19 inch diagonal width
 - 1280 x 1024 dot resolution
 - Horizontal & Vertical viewing angle: 176 degrees
 - Horizontal synchronization range: 31.5 - 81.1 kHz
 - Vertical synchronization range: 50 - 85 kHz
 - new black bezel
- **Optional MOD - Magnetic Optical Disk Drive (MOD) drive:**
 - Erasable, rewritable media
 - 2.3 GB, 5.25 inch form factor
 - Assigned to DICOM format image file
 - Stores 4,700 lossless JPEG compressed 512x512 image files/side
 - Off-line retrieval of image. Images may be viewed as soon as they are restored from MOD
- **PT – Peripheral Tower**
 - **DVD-RAM (Archive or Interchange)**
 - 9.4 GB total.
 - Up to 7,168 image storage
 - 16x DVD, 40x CD Max
 - 5.25" cartridge form factor
 - Xstream HD supports DVD-RAM, CD-R, CD-RW, DVD-R, DVD-RW
 - **DVD-R (Interchange):**
 - 9.4 GB total, 5.25" half height form factor
 - 20x DVD, 48x CD Max
 - Supports CD-R, CD-RW, DVD-R, DVD-RW

7. MAINTENANCE READINESS – SPARE PARTS

Below are the new FRUs for the GOC6 Operator Console.

Note: Always refer to the most current revision of documentation.

Part Number	DETAILED PART DESCRIPTION
GOC6 Operator Console 5212920-300	Complete LightSpeed VCT (64) 7.1 GOC6 Operator Console
5255299-20	VDIG
5255299-30	VeRB (Future Option - Vision Release)
5199748	High Speed Disk Array
5266222	Console intercom prescribed tilt hub - GOC6
5263798	Console Network Switch
5306873	Cable, SCIM, GOC6
5306867	Cable, RHARD, GOC6
5306870	Cable, X-ray Abort, GOC6
5306871	Cable, Audio-In
5306872	Cable, Audio-Out
5306478	Cable, J20 GOC6
5307778	Cable, SAS, 1 Meter
5313766	Cable, Ethernet CAT5e, 1 Meter
5313765	Cable, Ethernet CAT5e, 1.5 Meter
5314704	Cable, Power IEC C13-C14, 2 Meters
5316488	Cable, USB 2.0 type A - A, 0.5 Meter
5316994	Side Covers Collector, GOC6
5267233	front cover asm, console
5272833	outlet box asm GOC6
5316413	Cable, console power switch GOC6
5122589	Adapter, Bulkhead RJ45 CAT5e Shielded, L-com #TDG1026KS-C5E
5122507	Bulkhead Adapter, Fiber Optic - LC to LC Duplex, plastic body, ceramic sleeve, 0.525in sq. panel opening: LCOM FOA-550A-OD1
5128594	Connector, filtered sub-d 25position adapter
5151493	USB A BULKHEAD RECEPTACLE
5312224	Monitor table top asm, GOC6
5311788	Keyboard table top, GOC6
5124071	Fan, Tube Axial - 115VAC, 50-60Hz, 119mm x 119mm x 25mm x 95-115cfm, ball bearing motor Sinwan S125AP-11-1
5307276	Cable, Power, Fan Assembly for GOC6
5267534	Rear cover asm GOC6 console
2383761	USB Service Keys
5316032	Switch, 1 position with handles
5308763	RAID Controller with 512MB RAM, ASIC400, 4xSAS Host Ports and 1xSAS Expansion Port Compatible with 5308759

5316024	Dual Speed 3000RPM and 5300RPM Fan Module compatible with 5308759
5308759	2U 12Bay SAS to SAS-SATA RAID Rack -RAID6 Capable- including the following components - Single controller with 512MB RAM, ASIC400, 4xSAS Host Ports, 1xSAS Expansion Port, Dual Hot-swappable 460W PS, Redundant Dual-speed Cooling Modules
5308760	HDD w/ trays
5316023	480W Hot Swappable Power Supply compatible with 5308759
5308762	Trays w/o HDD
2382000	Trackball - USB
5270510	Monitor - 19" LCD
5305923	Peripheral Media Tower, 1 Bay MOD Enclosure (Option)
5183547-14	USB Mouse
5316029	SCIM
5169069-2	Monitor 19"
5315013-2	1.5 M Power Cable
5271574-2	Barcode Reader
5315370	CABLE, USB TYPE A-B
5316673	Cable, USB 2.0 type A - A, 2 Meter
5316577	Cable, Fiber Optic 50 125, GOC6 HDIG to Bulkhead
5183547-38	GEHC MI and CT Integrated Buy Level PC Workstation. HP8400 with 4GB Memory
5305658-2-JA	CT SCIM Overlay, Japanese, Silver for GOC6
5305658-2-EN	CT SCIM Overlay, English, Silver for GOC6
5305658-2-DA	CT SCIM Overlay, Danish, Silver for GOC6
5305658-2-NL	CT SCIM Overlay, Dutch Netherlands, Silver for GOC6
5305658-2-FR	CT SCIM Overlay, French, Silver for GOC6
5305658-2-DE	CT SCIM Overlay, German, Silver for GOC6
5305658-2-IT	CT SCIM Overlay, Italian, Silver for GOC6
5305658-2-NO	CT SCIM Overlay, Norwegian, Silver for GOC6
5305658-2-ES	CT SCIM Overlay, Spanish, Silver for GOC6
5305658-2-SV	CT SCIM Overlay, Swedish, Silver for GOC6
5305658-2-PT-PT	CT SCIM Overlay, Portuguese European, Silver for GOC6
5305658-2-PT-BR	CT SCIM Overlay, Portuguese Brazilian, Silver for GOC6
5305658-2-PL	CT SCIM Overlay, Polish, Silver for GOC6
5305658-2-SK	CT SCIM Overlay, Slovak, Silver for GOC6
5305658-2-TR	CT SCIM Overlay, Turkish, Silver for GOC6
5305658-2-LT	CT SCIM Overlay, Lithuanian, Silver for GOC6
5305658-2-FI	CT SCIM Overlay, Finnish, Silver for GOC6
5305658-2-HU	CT SCIM Overlay, Hungarian, Silver for GOC6
5305658-2-RO	CT SCIM Overlay, Romanian, Silver for GOC6
5305658-2-ET	CT SCIM Overlay, Estonian, Silver for GOC6
5305658-2-ZH	CT SCIM Overlay, Chinese, Silver for GOC6
5324605-EN	Keyboard Collector, GOC6, English
5324605-DA	Keyboard Collector, GOC6, Danish
5324605-NL	Keyboard Collector, GOC6, Dutch
5324605-FR	Keyboard Collector, GOC6, French
5324605-DE	Keyboard Collector, GOC6, German

5324605-IT	Keyboard Collector, GOC6, Italian
5324605-NO	Keyboard Collector, GOC6, Norwegian
5324605-ES	Keyboard Collector, GOC6, Spanish
5324605-SV	Keyboard Collector, GOC6, Swedish
5324605-PT-PT	Keyboard Collector, GOC6, Portuguese European
5324605-PT-BR	Keyboard Collector, GOC6, Portuguese Brazilian

8. SERVICE TOOLS

**Note: The USB service key is currently not compatible for use between MR and CT.
This issue will be addressed in the future.**

No new Service Tools.

The following service software disks are tab distributed:

5318761 – REV 1 Version 08MW33C.2
LightSpeed 7.1 Class C for GOC6 Consoles
VCT 64, VCT XT & VCT Std

5317906 – REV 1 Version 08MW33M.2
LightSpeed 7.1 Class M for GOC6 Consoles
VCT 64, VCT XT & VCT Std

8.1 SERVICE INFORMATION

- Early GOC6 installs will come with a beige PS2 keyboard instead of a black USB keyboard. The USB keyboards won't be available until the end of Q1, so for PS2 keyboard installation follow service note SN08100112a.
- Because the early systems will be using a PS2 keyboard the systems will display a USB port error when the system is shutdown, these errors only occur during shutdown and do not affect the system functionality or performance during system operation and can be ignored this issue can also be caused by PS2 mouse instead of a USB mouse.
- The USB mouse cannot be unplugged while applications are up because the mouse will no longer function, forcing the Technologist to reboot the system via a power recycle. A software fix for this issue is being investigated and will be out on the next software release. This affects all GOC6 console based systems.
- If a drive fails in the disk array the drive should be removed as soon as possible (even if it cannot be replaced) so that the bad drive will not cause unwanted side effects that may lead to scan aborts. The array is fault tolerant and can function with one less drive; the customer can continue to scan without data loss. To prevent downtime due to a 2nd drive failure, the drive should be replaced as soon as possible.

9. DOCUMENTATION

All new documentation is provided for this scanner. The list below identifies most of the manuals for this program.

Always refer to the most current revision of documentation.

9.1 VCT LIGHTSPEED GOC6 SERVICE DOCUMENTATION LIST

DIRECTION	TITLE	Paper	CD / DVD Media	DISTRIBUTION
5318716-2EN • 5318716-8EN (Web)	GOC6 Service Information Class A		X	Shipped with system
5321294-2EN • 5321294-8EN (Web)	GOC6 Service Information Class C		X	CT DVD via TAB
5116411-100	Installation Manual	X		Shipped with system
5116416-100	Illustrated Parts Manual	X		Shipped with system

10. SITE PREPARATION

It is very important to review and follow the siting specification in the Pre-installation manual.

11. FIELD ENGINEER TRAINING - CBT

The course code is CT6020 and all Field Engineers must have LightSpeed VCT training before reviewing LightSpeed VCT GOC6 Computer Based Training in MyLearning. LightSpeed VCT classes are available at the Healthcare institute.

12. SERVICE FEATURES

- RAC Diagnostics are no longer used (no physical RAC)
- gre-raid Diagnostics scripts have been modified for use with the GOC6 software/hardware
- ConsoleDiags Class C scripts have been modified for use with the GOC6 software/hardware

13. SPRS FIXED WITH FDR2 (08MW33.2) SOFTWARE

List of SPR's Resolved between GOC5 and GOC6 (08MW33.2)

- None.

14. PLANNED MAINTENANCE

PM strategy for the GOC6 Operator Console will be the same as the GOC5 Operator Console.